Analysis report

Holzinger–Swineford · MLR · listwise · N = 156 + 145 · lavaan 0.7.3

Table 1

Parameter estimates

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
Pasteur	visual	=~	x1		1.000	0.000	—	—	1.000	1.000	0.887
Pasteur	visual	=~	x2		0.394	0.197	1.996	.046	0.007	0.780	0.336
Pasteur	visual	=~	x3		0.570	0.201	2.838	.005	0.176	0.964	0.515
Pasteur	textual	=~	x4		1.000	0.000	—	—	1.000	1.000	0.823
Pasteur	textual	=~	x5		1.183	0.106	11.213	< .001	0.977	1.390	0.856
Pasteur	textual	=~	x6		0.875	0.093	9.388	< .001	0.692	1.058	0.838
Pasteur	speed	=~	x7		1.000	0.000	—	—	1.000	1.000	0.547
Pasteur	speed	=~	x8		1.125	0.262	4.298	< .001	0.612	1.638	0.682
Pasteur	speed	=~	x9		0.922	0.297	3.099	.002	0.339	1.505	0.551
Pasteur	x1	~~	x1		0.298	0.338	0.882	.378	-0.365	0.961	0.214
Pasteur	x2	~~	x2		1.334	0.179	7.441	< .001	0.982	1.685	0.887
Pasteur	x3	~~	x3		0.989	0.154	6.444	< .001	0.689	1.290	0.735
Pasteur	x4	~~	x4		0.425	0.073	5.867	< .001	0.283	0.567	0.322
Pasteur	x5	~~	x5		0.456	0.086	5.304	< .001	0.287	0.624	0.267

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
Pasteur	x6	~~	x6		0.290	0.055	5.267	< .001	0.182	0.398	0.297
Pasteur	x7	~~	x7		0.820	0.132	6.211	< .001	0.561	1.079	0.701
Pasteur	x8	~~	x8		0.510	0.101	5.048	< .001	0.312	0.708	0.535
Pasteur	x9	~~	x9		0.680	0.126	5.406	< .001	0.434	0.927	0.696
Pasteur	visual	~~	visual		1.097	0.378	2.899	.004	0.355	1.839	1.000
Pasteur	textual	~~	textual		0.894	0.160	5.577	< .001	0.580	1.209	1.000
Pasteur	speed	~~	speed		0.350	0.140	2.502	.012	0.076	0.623	1.000
Pasteur	visual	~~	textual		0.479	0.144	3.334	< .001	0.197	0.761	0.484
Pasteur	visual	~~	speed		0.185	0.076	2.449	.014	0.037	0.333	0.299
Pasteur	textual	~~	speed		0.182	0.076	2.402	.016	0.033	0.330	0.325
Pasteur	x1	~1			4.941	0.095	52.249	< .001	4.756	5.127	4.183
Pasteur	x2	~1			5.984	0.098	60.949	< .001	5.792	6.176	4.880
Pasteur	x3	~1			2.487	0.093	26.778	< .001	2.305	2.669	2.144
Pasteur	x4	~1			2.823	0.092	30.689	< .001	2.642	3.003	2.457
Pasteur	x5	~1			3.995	0.105	38.183	< .001	3.790	4.200	3.057
Pasteur	x6	~1			1.922	0.079	24.321	< .001	1.767	2.077	1.947

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
Pasteur	x7	~1			4.432	0.087	51.181	< .001	4.263	4.602	4.098
Pasteur	x8	~1			5.563	0.078	71.214	< .001	5.410	5.716	5.702
Pasteur	x9	~1			5.418	0.079	68.440	< .001	5.263	5.573	5.480
Pasteur	visual	~1			0.000	0.000	—	—	0.000	0.000	0.000
Pasteur	textual	~1			0.000	0.000	—	—	0.000	0.000	0.000
Pasteur	speed	~1			0.000	0.000	—	—	0.000	0.000	0.000
Grant-White	visual	=~	x1		1.000	0.000	—	—	1.000	1.000	0.677
Grant-White	visual	=~	x2		0.736	0.190	3.870	< .001	0.363	1.109	0.517
Grant-White	visual	=~	x3		0.925	0.216	4.281	< .001	0.501	1.348	0.694
Grant-White	textual	=~	x4		1.000	0.000	—	—	1.000	1.000	0.866
Grant-White	textual	=~	x5		0.990	0.089	11.179	< .001	0.816	1.163	0.829
Grant-White	textual	=~	x6		0.963	0.091	10.588	< .001	0.785	1.142	0.826
Grant-White	speed	=~	x7		1.000	0.000	—	—	1.000	1.000	0.659
Grant-White	speed	=~	x8		1.226	0.162	7.560	< .001	0.908	1.544	0.796
Grant-White	speed	=~	x9		1.058	0.250	4.227	< .001	0.567	1.548	0.701
Grant-White	x1	~~	x1		0.715	0.182	3.938	< .001	0.359	1.071	0.542

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
Grant-White	x2	~~	x2		0.899	0.142	6.338	< .001	0.621	1.177	0.733
Grant-White	x3	~~	x3		0.557	0.122	4.568	< .001	0.318	0.796	0.519
Grant-White	x4	~~	x4		0.315	0.066	4.764	< .001	0.186	0.445	0.251
Grant-White	x5	~~	x5		0.419	0.071	5.876	< .001	0.279	0.559	0.312
Grant-White	x6	~~	x6		0.406	0.076	5.363	< .001	0.258	0.554	0.317
Grant-White	x7	~~	x7		0.600	0.100	5.995	< .001	0.404	0.797	0.566
Grant-White	x8	~~	x8		0.401	0.159	2.516	.012	0.089	0.714	0.367
Grant-White	x9	~~	x9		0.535	0.140	3.814	< .001	0.260	0.810	0.509
Grant-White	visual	~~	visual		0.604	0.194	3.105	.002	0.223	0.985	1.000
Grant-White	textual	~~	textual		0.942	0.162	5.807	< .001	0.624	1.260	1.000
Grant-White	speed	~~	speed		0.461	0.120	3.839	< .001	0.226	0.697	1.000
Grant-White	visual	~~	textual		0.408	0.115	3.535	< .001	0.182	0.634	0.541
Grant-White	visual	~~	speed		0.276	0.101	2.731	.006	0.078	0.474	0.523
Grant-White	textual	~~	speed		0.222	0.105	2.115	.034	0.016	0.427	0.336
Grant-White	x1	~1			4.930	0.095	51.696	< .001	4.743	5.117	4.293
Grant-White	x2	~1			6.200	0.092	67.416	< .001	6.020	6.380	5.599

Group	Left	Relation	Right	Label / value	Estimate	SE	z	p	95% CI lower	95% CI upper	std.all
Grant-White	x3	~1			1.996	0.086	23.195	< .001	1.827	2.164	1.926
Grant-White	x4	~1			3.317	0.093	35.625	< .001	3.135	3.500	2.959
Grant-White	x5	~1			4.712	0.096	48.986	< .001	4.524	4.901	4.068
Grant-White	x6	~1			2.469	0.094	26.277	< .001	2.285	2.653	2.182
Grant-White	x7	~1			3.921	0.086	45.819	< .001	3.753	4.089	3.805
Grant-White	x8	~1			5.488	0.087	63.174	< .001	5.318	5.659	5.246
Grant-White	x9	~1			5.327	0.085	62.571	< .001	5.160	5.494	5.196
Grant-White	visual	~1			0.000	0.000	—	—	0.000	0.000	0.000
Grant-White	textual	~1			0.000	0.000	—	—	0.000	0.000	0.000
Grant-White	speed	~1			0.000	0.000	—	—	0.000	0.000	0.000

Note. Estimates and intervals are unstandardised; Std. all is fully standardised. Values are rounded only for display. A dash indicates unavailable output. 95% CI.

Table 2

Fit measures

Measure	Value
χ^2	115.851
df	48.000
p (χ^2)	< .001
CFI	0.923
TLI	0.885
RMSEA	0.097
RMSEA 90% CI lower	0.075
RMSEA 90% CI upper	0.120
SRMR	0.068
Scaled χ^2	121.741
Scaled df	48.000
p (scaled χ^2)	< .001
Robust CFI	0.921
Robust TLI	0.881
Robust RMSEA	0.099
Robust RMSEA 90% CI lower	0.077
Robust RMSEA 90% CI upper	0.120
Scaled CFI	0.914
Scaled TLI	0.872
Scaled RMSEA	0.101
Scaled RMSEA 90% CI lower	0.078
Scaled RMSEA 90% CI upper	0.124

Note. Scaled and robust statistics are labelled explicitly. Compare the appropriate version for your estimator.

Table 3

Explained variance (R^2)

Group	Left	R^2
Pasteur	x1	0.786
Pasteur	x2	0.113
Pasteur	x3	0.265
Pasteur	x4	0.678
Pasteur	x5	0.733
Pasteur	x6	0.703
Pasteur	x7	0.299
Pasteur	x8	0.465
Pasteur	x9	0.304
Grant-White	x1	0.458
Grant-White	x2	0.267
Grant-White	x3	0.481
Grant-White	x4	0.749
Grant-White	x5	0.688
Grant-White	x6	0.683
Grant-White	x7	0.434
Grant-White	x8	0.633
Grant-White	x9	0.491

Table 4

Measurement invariance

Stage	χ^2	df	p	CFI	RMSEA	SRMR	ΔCFI	ΔRMSEA	$\Delta\chi^2$	Δdf	Difference p	CFI variant	RMSEA variant
Configural	121.741	48	< .001	0.921	0.099	0.068	—	—	—	—	—	cfi.robust	rmsea.robust
Metric	125.997	54	< .001	0.920	0.093	0.072	-0.001	-0.005	6.567	6	.363	cfi.robust	rmsea.robust
Scalar	166.748	60	< .001	0.881	0.108	0.082	-0.039	0.014	40.838	6	< .001	cfi.robust	rmsea.robust
Strict	181.429	69	< .001	0.873	0.104	0.088	-0.008	-0.004	15.694	9	.074	cfi.robust	rmsea.robust

Note. Each stage is compared with the previous stage using lavaan's difference test. ΔCFI and ΔRMSEA are signed changes; the exact fit-index variants are reported. These statistics do not automatically establish invariance. Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001") Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001") Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")

Table 5

Modification indices

Group	Left	Relation	Right	MI	EPC	Standardised EPC
Grant-White	x7	~~	x8	24.819	0.612	1.247
Grant-White	visual	=~	x9	24.539	0.748	0.567
Pasteur	x4	~~	x6	11.280	-0.326	-0.928
Grant-White	visual	=~	x7	11.267	-0.504	-0.380
Pasteur	visual	=~	x9	11.073	0.304	0.322
Pasteur	textual	=~	x1	10.185	0.944	0.756
Pasteur	visual	=~	x5	9.593	-0.281	-0.225
Grant-White	textual	=~	x8	9.572	-0.290	-0.269
Grant-White	x1	~~	x9	9.332	0.205	0.332
Grant-White	textual	=~	x9	9.201	0.268	0.253
Pasteur	x2	~~	x3	8.961	0.329	0.286
Grant-White	x8	~~	x9	8.494	-0.398	-0.859
Pasteur	x4	~~	x7	7.197	0.159	0.270
Pasteur	x2	~~	x7	6.851	-0.241	-0.230
Pasteur	textual	=~	x3	6.203	-0.393	-0.320
Pasteur	x4	~~	x5	5.463	0.315	0.715
Grant-White	x1	~~	x7	5.236	-0.158	-0.241
Pasteur	visual	=~	x7	5.182	-0.227	-0.220
Pasteur	x3	~~	x9	4.526	0.158	0.192
Grant-White	x7	~~	x9	4.332	-0.208	-0.367
Grant-White	x4	~~	x8	4.201	-0.095	-0.268
Pasteur	x7	~~	x8	4.187	0.342	0.528
Pasteur	visual	=~	x6	3.972	0.137	0.146
Pasteur	x1	~~	x2	3.971	-0.340	-0.539

Group	Left	Relation	Right	MI	EPC	Standardised EPC
Pasteur	x1	~~	x4	3.932	0.119	0.334
Pasteur	x3	~~	x5	3.757	-0.134	-0.199
Grant-White	x3	~~	x7	3.534	-0.116	-0.201
Grant-White	x2	~~	x3	3.441	0.176	0.249
Pasteur	speed	=~	x5	3.333	-0.293	-0.132
Pasteur	x2	~~	x9	2.830	0.141	0.148
Grant-White	speed	=~	x1	2.780	0.361	0.214
Grant-White	visual	=~	x8	2.749	-0.273	-0.203
Pasteur	speed	=~	x6	2.661	0.200	0.120
Grant-White	speed	=~	x5	2.579	0.189	0.111
Pasteur	x2	~~	x4	2.275	-0.107	-0.141
Pasteur	x1	~~	x5	2.240	-0.100	-0.271
Pasteur	x5	~~	x7	1.914	-0.090	-0.147
Pasteur	x3	~~	x7	1.724	-0.107	-0.118
Pasteur	x1	~~	x9	1.595	0.091	0.203
Pasteur	x6	~~	x8	1.566	0.055	0.143
Pasteur	x7	~~	x9	1.543	-0.157	-0.210
Pasteur	visual	=~	x4	1.498	0.099	0.090
Grant-White	x1	~~	x3	1.468	-0.164	-0.260
Pasteur	x4	~~	x8	1.349	-0.061	-0.130
Pasteur	x3	~~	x6	1.345	0.062	0.115
Grant-White	x3	~~	x6	1.314	0.061	0.129
Pasteur	x5	~~	x6	1.303	0.138	0.380
Pasteur	x4	~~	x9	1.290	-0.062	-0.114
Pasteur	textual	=~	x8	1.176	-0.120	-0.116

Group	Left	Relation	Right	MI	EPC	Standardised EPC
Grant-White	x2	~~	x5	1.146	-0.067	-0.109
Grant-White	speed	=~	x3	1.071	-0.206	-0.135
Grant-White	x3	~~	x9	1.047	0.062	0.113
Pasteur	x2	~~	x6	1.007	0.060	0.096
Pasteur	visual	=~	x8	0.925	-0.097	-0.104
Pasteur	x1	~~	x7	0.804	-0.071	-0.143
Grant-White	textual	=~	x2	0.741	-0.111	-0.098
Grant-White	x3	~~	x5	0.699	-0.046	-0.095
Grant-White	speed	=~	x4	0.691	-0.093	-0.056
Grant-White	x5	~~	x7	0.687	0.044	0.087
Pasteur	x1	~~	x3	0.641	-0.240	-0.442
Pasteur	x1	~~	x8	0.628	-0.059	-0.150
Grant-White	x5	~~	x9	0.587	0.039	0.083
Pasteur	speed	=~	x2	0.587	-0.168	-0.081
Grant-White	speed	=~	x2	0.569	-0.145	-0.089
Pasteur	x3	~~	x4	0.546	-0.046	-0.072
Grant-White	speed	=~	x6	0.525	-0.083	-0.050
Pasteur	speed	=~	x3	0.522	0.149	0.076
Grant-White	x4	~~	x9	0.510	0.034	0.083
Pasteur	x8	~~	x9	0.498	-0.109	-0.185
Grant-White	x4	~~	x7	0.493	0.034	0.079
Grant-White	x1	~~	x2	0.493	-0.073	-0.090
Pasteur	textual	=~	x7	0.486	0.076	0.066
Grant-White	x2	~~	x7	0.446	-0.047	-0.064
Grant-White	x2	~~	x4	0.439	0.039	0.072

Group	Left	Relation	Right	MI	EPC	Standardised EPC
Pasteur	textual	=~	x2	0.388	-0.086	-0.066
Pasteur	x1	~~	x6	0.361	0.031	0.104
Pasteur	x5	~~	x8	0.337	-0.033	-0.069
Pasteur	x3	~~	x8	0.321	0.040	0.057
Grant-White	x5	~~	x8	0.317	0.028	0.069
Grant-White	textual	=~	x1	0.310	0.083	0.070
Pasteur	textual	=~	x9	0.247	0.049	0.047
Grant-White	x6	~~	x9	0.222	-0.024	-0.051
Grant-White	visual	=~	x5	0.199	-0.058	-0.039
Pasteur	x5	~~	x9	0.199	0.026	0.048
Grant-White	x1	~~	x8	0.173	0.028	0.052
Grant-White	x1	~~	x4	0.164	0.023	0.049
Grant-White	x6	~~	x8	0.140	-0.018	-0.045
Grant-White	textual	=~	x7	0.137	0.033	0.031
Grant-White	visual	=~	x6	0.083	0.037	0.025
Pasteur	speed	=~	x4	0.068	0.038	0.019
Grant-White	x2	~~	x6	0.061	-0.015	-0.025
Pasteur	speed	=~	x1	0.041	-0.066	-0.033
Pasteur	x6	~~	x7	0.035	0.009	0.019
Grant-White	textual	=~	x3	0.029	0.023	0.022
Grant-White	x3	~~	x8	0.028	-0.010	-0.021
Grant-White	visual	=~	x4	0.023	0.019	0.013
Grant-White	x1	~~	x6	0.020	-0.009	-0.016
Grant-White	x3	~~	x4	0.020	-0.007	-0.017
Grant-White	x4	~~	x5	0.019	0.017	0.048

Group	Left	Relation	Right	MI	EPC	Standardised EPC
Grant-White	x4	~~	x6	0.018	-0.016	-0.045
Grant-White	x6	~~	x7	0.016	-0.007	-0.013
Grant-White	x2	~~	x8	0.014	-0.008	-0.013
Pasteur	x2	~~	x5	0.008	0.007	0.009
Pasteur	x6	~~	x9	0.008	-0.004	-0.009
Grant-White	x2	~~	x9	0.004	0.004	0.006
Pasteur	x2	~~	x8	0.002	0.004	0.004
Grant-White	x1	~~	x5	0.001	-0.001	-0.003
Grant-White	x5	~~	x6	0.000	-0.000	-0.001

Note. Modification indices concern fixed-to-zero parameters, not equality constraints. Add a theoretically justified parameter, then rerun the model. An addition applies only to the indicated group.

Table 6

Reliability & validity

Group	Factor	Composite ω	AVE
Pasteur	visual	0.584	0.382
Pasteur	textual	0.877	0.707
Pasteur	speed	0.620	0.351
Grant-White	visual	0.661	0.400
Grant-White	textual	0.878	0.706
Grant-White	speed	0.766	0.521

Note. Unit-weight omega-type composite reliability and AVE use observed variances (semTools). HTMT2 uses absolute correlations and geometric means; ordinal indicators use polychoric correlations. Reverse-code items in the source data when appropriate. semTools 0.5.10.

Table 7

Discriminant validity

Group	Factor 1	Factor 2	HTMT2	Latent r	$\sqrt{AVE_1}$	$\sqrt{AVE_2}$	Fornell–Larcker
Pasteur	visual	textual	0.313	0.484	0.618	0.841	Yes
Pasteur	visual	speed	0.292	0.299	0.618	0.593	Yes
Pasteur	textual	speed	0.337	0.325	0.841	0.593	Yes
Grant-White	visual	textual	0.513	0.541	0.632	0.840	Yes
Grant-White	visual	speed	0.420	0.523	0.632	0.722	Yes
Grant-White	textual	speed	0.347	0.336	0.840	0.722	Yes

Note. Fornell–Larcker compares each factor's square root of AVE with the absolute fitted latent correlation. “Yes” means both square roots exceed that correlation; it is not a general validity verdict.